Title: Muscle coactivation levels reflect a trade-off between metabolic cost and performance when responding to physical disturbances during upper limb posture control.

Abbreviated Title: Postural Coactivation Shows Cost-Performance Trade-off

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Abstract

Muscle coactivation refers to the simultaneous activity of muscles with opposing functions. Recent observations suggest that coactivation improves the nervous system's ability to respond to sensory feedback. However, muscle activity requires energy, and the relationship between this energy use and performance is not fully understood. This study investigated the trade-off between metabolic cost and performance under different levels of muscle coactivation. Healthy participants attempted to maintain a fixed arm posture in a Kinarm exoskeleton robot while encountering mechanical disturbances. The disturbances rapidly extended or flexed the elbow and displaced the participant's arm from the target, requiring them to return to the target. Participants first performed the task under self-selected levels of muscle coactivation, followed by biofeedback to adjust coactivation to [0.5, 2, 4, 8, 16]x their self-selected (1x) level. Metabolic cost was measured with gas respirometry, muscle activity using surface EMG, and behavioral responses using success rates. Trials were considered successful if participants returned their arm back to the target within 400 milliseconds (ms) post-perturbation. Pilot data suggest that at self-selected levels of muscle coactivation, participants achieved a mean success rate of 54%. Higher levels of coactivation led to a reduction in peak arm displacement following the same disturbances, as well as increased muscle responsiveness and success rates were nearly 100%. However, increased coactivation levels resulted in an increased metabolic cost. These findings suggest that participants may prioritize metabolic cost over performance at self-selected coactivation levels. This research provides insights into the role of muscle coactivation in sensory feedback response.

Introduction

Muscle coactivation refers to the simultaneous activation of pairs of agonist and antagonist muscles (Vieira et al., 2019). Coactivation influences a diverse range of actions, from balance to precision movement (Vieira et al., 2019). A longstanding and impactful idea in systems neuroscience is that coactivation stabilizes the body by making muscles stiff and resistant to motion to minimize errors caused by physical perturbations (Vieira et al., 2019). However, new research indicates that muscle coactivation may not only stiffen muscles but also improve the nervous system's responsiveness to sensory input (Balshaw et al., 2018). Sprinters, for example, coactivate their muscles while waiting in the blocks for the starting gun, which seems counterintuitive, given that the goal is to start and finish the race as quickly as possible. This highlights a gap in understanding the role muscle coactivation plays in human movement.

Most of the current research on coactivation is based on isolated muscle experiments or preclinical models. In these investigations, scientists use calcium solutions or nerve stimulation to control muscle activity (Gebel et al., 2019). Researchers adjust calcium levels to change muscle activity and then apply a consistent stretch to measure displacement (Gebel et al., 2019). Using the equation $F = -kx$, where F is force, x is displacement, and k represents stiffness, they calculate stiffness as $F/x = -k$ (Latash, 2018). When muscles are more active, the displacement tends to be smaller for the same force, which means that k, the stiffness, is larger (Gebel et al., 2019). The studies offer valuable knowledge, but they do not track neural feedback that people experience when performing real-world activities, which need motor control and sensory processing (Kesar et al., 2019). These experiments lack a complete understanding of coactivation effects on movement because they do not track neural feedback.

A key gap in the literature is the limited focus on sensory feedback, which demands more attention. Many coactivation studies, especially those using surface electromyography (EMG), measure muscle activity but often skip analyzing reflexive responses to perturbations or how sensory input is processed. Instead, they focus on stiffness, estimated from the relationship $F = -kx$ (Latash, 2018). These studies typically lump displacement estimates over 300–500 ms, mixing in reflexive and neural contributions without isolating them (Balshaw et al., 2018). Coactivation can reduce movement variability and boost precision (Balshaw et al., 2018), but it's unclear how it shapes the nervous system's ability to handle sensory data during disturbances. The short-latency response (SLR), a fast spinal reflex, fires within 25–30 ms after a muscle stretch, likely driven by coactivation's effect on muscle sensitivity (Maurus et al., 2024). Then, the long-latency response (LLR) kicks in at 50–100 ms, to provide adaptable motor control through cortical processing, with coactivation possibly shaping its adaptability (Maurus et al., 2024). Since sensory feedback drives precise movement control, we need to dig deeper into whether higher coactivation alters the neural processing of these signals. Another underexplored area is the trade-off between metabolic cost and performance. While enhanced coactivation improves motor control and responsiveness according to Peterson & Martin (2010), it increases energy requirements which may reduce endurance for long activities.

This study examines how muscle coactivation affects the trade-off between performance and metabolic cost during upper limb posture control with mechanical disturbances, an area requiring further research. The hypothesis is that there is a trade-off between muscle coactivation, metabolic cost, and task performance. Higher coactivation levels may improve responsiveness to perturbations and improve performance (lower error rates), but it might come at the expense of an increased metabolic cost. On the contrary, lower coactivation may save energy expenditure but reduce performance. Coactivation enhances stability and control, yet its energy impact in rapid, precise movements is unclear. By exploring this balance, the research seeks to provide insight into how the nervous system adjusts coactivation for optimal stability and energy efficiency, with findings to inform future analyses and understand motor control.

Methodology

Participant Characteristics

Twelve right-handed participants (7 male and 5 female) were recruited from the University of Calgary. The age, height, and weight of the participants were 21.92 ± 5.96 years, 173.17 ± 11.26 cm and 74.16 ± 17.80 kg, respectively (mean ± SD). Only participants who were free of self-reported neurological or musculoskeletal conditions were included in the study, and participants had normal or corrected vision. All participants signed consent forms before doing the experiment, and the study protocol was approved by the University of Calgary Conjoint Health Research Ethics Board (CHREB). The entire data collection session lasted 1.5-2 hours, including apparatus set-up. Each participant was paid $20 CAD at the end to compensate for travel and time expenses.

Apparatus set-up

The Kinarm exoskeleton robot (Calgary, AB, Canada) was used as the primary apparatus for this experiment, designed to restrict arm movements to a horizontal plane. Participants' arms were placed into the troughs of the device, which permitted smooth lateral motion while preventing vertical displacement. Adjustments were made to the chair height to align participants' arms with the exoskeleton, ensuring feet rested flat on the footrest and backs were supported comfortably. A shoulder indicator light ensured shoulders remained level, aligned with the acromion process, while armrests were tuned to hold arms comfortably with elbows slightly bent. The robot's joints were synced to match participants' shoulder, elbow, and wrist positions. The exoskeleton enabled nearly frictionless movement at these joints and could apply controlled forces to the shoulder or elbow while tracking motion. Targets (0.75 cm in size) and a white cursor (0.5 cm radius), tied to the index finger, were shown on a TV screen connected to the Kinarm. A shield was used to block direct vision of the arm and hand.

The Cosmed K5 wearable metabolic system was also utilized to measure metabolic activity. The K5 mask, covering the nose and mouth, was fitted snugly, and participants were asked to limit talking to avoid skewing breathing data. This system captured VO_2 and VCO_2, with start and end times of each measurement period (resting and the six conditions) recorded from the Cosmed computer. These time points were used in the analysis of metabolic cost using the Brockway equation (1987), which converts VO_2 into energy expenditure (measured in Watts).

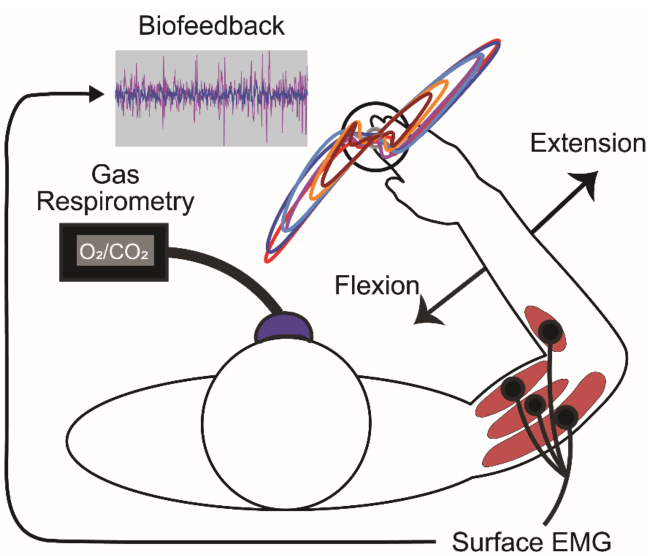

Figure 1. Schematic of the task protocol setup. This figure illustrates the experimental setup for the task protocol, depicting a side view of the participant's arm with the elbow positioned for extension (outward arrow) and flexion (inward arrow) movements induced by the Kinarm exoskeleton robot. Concentric circles at the right hand represent exemplar arm trajectories for each coactivation level. A biofeedback signal, shown as a wavy line inset, indicates the real-time muscle activation feedback used to adjust coactivation to the specified levels. The VO_2 mask, colored blue, is positioned over the participant's face to measure metabolic cost via gas respirometry, while red areas on the arm highlight the four muscles (brachioradialis, biceps, triceps lateralis, and triceps longus) monitored using surface EMG.

Task Protocol

The experiment was structured into three distinct tasks, each with specific procedures outlined below.

Task 1: Self-Selected Coactivation Condition with Mechanical Disturbances

Before starting, baseline resting breathing rates were recorded after participants rested for five minutes. The starting and ending times of the breathing rate were noted in a separate Excel sheet to capture the baseline metabolic cost, with VO_2 and VCO_2 measured using the Cosmed K5. Testing for the participant began by initiating the cardiopulmonary exercise testing (CPET) and breath-by-breath measure in the database, which allowed continuous measurement of VO_2 and VCO_2 throughout all conditions. The initial time was recorded from the COSMED computer to mark the start of the baseline measurement, and the ending time was similarly recorded to determine when the baseline period ended. After the resting breathing rate measurement was collected, the first task, labelled condition 1x (self-selected), began with a small circular target at the screen's center. Participants guided the white cursor into the target and maintained a steady arm posture, holding the cursor within the target for 2000 ± 1000 ms. The Kinarm exoskeleton robot then introduced a mechanical disturbance on every trial, rapidly extending or flexing the elbow to displace the cursor from the target, as shown by the arrows in Figure 1. These disturbances were a torque (the rotational equivalent of force, calculated as Torque = F x distance, where F is the force applied at a perpendicular distance from the center of rotation) of 5 Newton-meters (Nm) for 80 ms at both the elbow and shoulder, which resulted in elbow movement. Participants were required to return the cursor to the center of the circular target within 400 ms. If the return occurred within the time limit, the target turned green (successful trial). If the return was too slow, the target turned blue, signalling that the participant did not meet the time requirement for that trial (unsuccessful or error). Short breaks of about 1 second were provided between trials to prevent fatigue. Time durations for the K5 metabolic mask were recorded at the beginning and end of condition 1x and noted in a separate Excel datasheet.

Task 2: Adjusted Coactivation Conditions with Biofeedback and Increasing Intensity

Before starting task 2, participants were asked to relax for 5 minutes. In the meantime, EMG data saved from task 1 were exported into MATLAB. Then, baseline biceps EMG activity data, recorded before the perturbation onset, were extracted for every trial in the self-selected condition. These data were averaged across trials in MATLAB to get a normalization constant for each participant. This constant was then multiplied by the coactivation levels (0.5x, 2x, 4x, 8x, and 16x) to set the target EMG level in the biofeedback task. These values were entered into the task protocol (TP) under the reference EMG column by editing the TP table.

In task 2, a circular target appeared on the screen, and participants actively moved the cursor into the target, holding it there for 2000 ± 1000 ms to prepare for the trial. A magenta slider displayed the participant's current muscle activation, while a red biofeedback box indicated the required activation level for the condition. The biofeedback box was centred on the target EMG level, with tolerance levels set to ±30% of the target. The slider's position shows the amount of biceps muscle activation. Higher muscle activation raised the slider, and lower activation lowered it. When the slider aligned within the red rectangular box, indicating the correct level of biceps contraction for the condition (0.5x, 2x, 4x, 8x, or 16x relative to the self-selected 1x level), the biofeedback box turned green, signalling that the participant had generated the appropriate amount of EMG. Participants were required to hold the slider in the biofeedback target for 500 ms, and at this point, the robot applied a torque of 5 Nm for 80 ms at the same position as Task 1 (elbow and shoulder) to displace the arm from the target. If the participants returned the white cursor in the circular target before 400 ms, the target turned green (successful trial), and if the return took longer than 400 ms, then the target turned blue (error or failed trial). The task increased in difficulty over time, with conditions progressing from 0.5x to 16x, and testing continued until completion or until participants could no longer complete the trials due to fatigue or inability to maintain the required coactivation level. Time points for the start and end of each condition using the Cosmed K5 were logged and recorded in the Excel datasheet to track the duration of metabolic measurements.

Task 3: Normalization Task for Establishing Baseline Muscle Activity

The third task, referred to as the normalization task, involved holding the cursor in the center of a circular target for approximately two minutes while participants relaxed under a small load. This task was designed to establish a baseline for muscle activity under minimal effort, providing a reference point for normalizing EMG data across participants and conditions. Each condition in Tasks 1 and 2 consisted of 80 trials, totaling 480 trials per participant, though the number of completed trials varied due to differences in endurance. Task 3 included six trials per participant to ensure sufficient data for normalization purposes.

Kinematics

Movement data were collected from the Kinarm system to evaluate arm kinematics throughout the tasks. Elbow joint angles were captured at a sampling rate of 1 kHz and processed with a second-order, dual-pass Butterworth filter using a 30 Hz cutoff frequency to remove noise. The delta elbow angle, measured in degrees (deg), was calculated to determine how much the perturbation displaces the participant's arm relative to the participant's arm posture in the start target. Data processing in MATLAB R2019b gives the movement time, elbow angles, and peak elbow angle.

Muscle Activity

Surface EMG was used to determine muscle activity during the tasks. Prior to attaching electrodes, the skin was prepared by shaving with disposable razors if necessary and cleaned with alcohol swabs to enhance signal quality. Bipolar surface electrodes (DE 2.1 Single Differential Electrode) (Delsys Bagnoli, Delsys Inc, Natick, MA, USA) were positioned over the brachioradialis, triceps lateralis, biceps and triceps longus to record activity from these upper limb muscles. A reference electrode was placed on the lateral epicondyle, and the EMG system, connected to a computer, recorded data at a sampling rate of 1 kHz. Signals were amplified with a gain of 1000 and subjected to a band-pass filter (20–450 Hz, dual-pass, 3rd-order Butterworth) before full-wave rectification to isolate muscle activity. Normalization was achieved by having participants hold reference loads of ±1 Nm to flex and extend the elbow, maintained for 15 seconds in two directions (Task 3). These normalized EMG values were expressed in normalized units (nu), enabling consistent comparisons across participants by accounting for individual muscle strength differences. For analysis, EMG data from all trials were aligned to the onset of perturbations. Muscle activity was averaged for each muscle within defined time windows: SLR (25–50 ms) and LLR (50–100 ms) (Maurus et al., 2024). Only elbow monoarticular and biarticular muscles were looked at because the perturbations were designed such that they did not evoke shoulder motion within the defined time windows. To isolate the perturbation-evoked response (ΔEMG, in normalized units), the average activity from unperturbed trials was subtracted from that of perturbed trials. The ΔEMG was calculated for each trial, averaged per participant, and across participants for each condition. All EMG data were processed and analyzed using custom MATLAB R2019b scripts.

Data Analysis and Outcome Measures

The study gathered data for metabolic cost through gas respirometry, checked success rates and speed in dynamic tasks, and tracked muscle coactivation with EMG to see how these changed with different coactivation levels. Mean metabolic rates were determined from VO_2 and VCO_2 data using the Brockway equation (1987). Performance was quantified using the error rate, calculated as one minus the percentage of successful trials (error rate = 1 - % successful trials). Behavioural responses were modelled with an exponential fit, using the equation:

General model Exponential Fit (1 term):

$$f(x) = a * e^{b*x} \qquad \textbf{(Equation 1)}$$

In this equation, x represents the independent variable (coactivation level), and f(x) is the dependent variable (behavioral response). The exponential model, as described by Ting and Macpherson (2005), allows us to a make relationship by adjusting parameters ‘a’ and ‘b’ to reflect initial values and growth/decay rates, respectively. If b > 0, it indicates exponential growth, and if b < 0, it represents exponential decay. This model will enhance our understanding of motor control dynamics. Parameters were extracted in MATLAB to quantify behavior across conditions. Figures were also generated in MATLAB to visualize trends.

Results

Kinematic Responses Across Coactivation Levels

The elbow angle data were recorded during the extension and flexion tasks to capture movement across coactivation levels (0.5x, 1x, 2x, 4x, 8x, 16x).

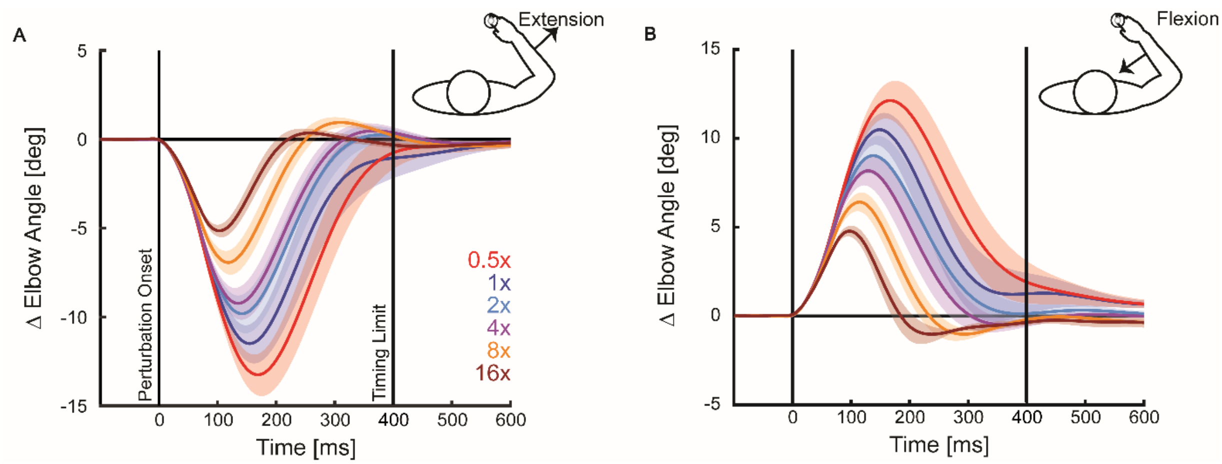

Figure 2. Average elbow angle of participants over time for the extension **(A)** and flexion tasks **(B)**. This provides a detailed view of elbow angle changes across all conditions. Each coactivation level is marked by a specific line color: 0.5x in light red, 1x in dark blue, 2x in light blue, 4x in purple, 8x in dark orange, and 16x in dark red. The angle starts at 0 degrees when the cursor is on the target and changes after the robot applies a disturbance. The flexion and extension patterns look similar, but the direction of movement is opposite. The shaded areas show the standard error of the trace across participants and and are widest at smaller coactivation levels (e.g., 0.5x) and narrowest at 16x..

Figures 2 and 3 illustrate that there is a systemic reduction in elbow angle with greater muscle coactivation. Looking at Figures 2A and 3A, the peak elbow angle reaches about -14 degrees at 0.5x (light red) and drops to around -5 degrees at 16x (dark red). Similarly, we can see that Figures 2B and 3B show smaller displacement from the target with higher coactivation levels (e.g., 8x, 16x), while the angle peaks at about 13 degrees for 0.5x and falls to around 5 degrees for 16x. The elbow moves the most when coactivation is low and the least when coactivation is high. Figures 2A and 2B also show that it takes less time to reach back into the center of the circular target with greater muscle coactivation.

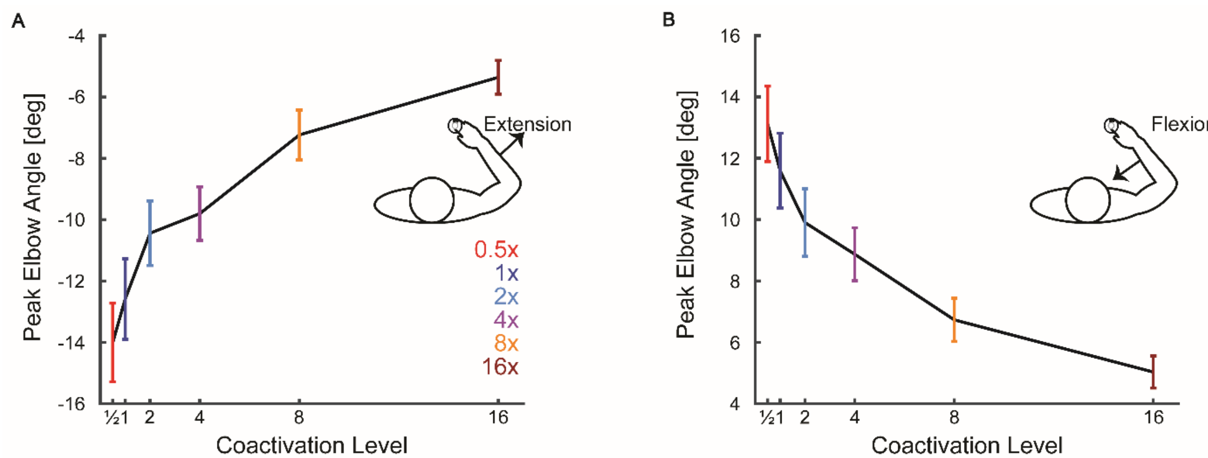

Figure 3. Summary of peak elbow angle for extension **(A)** and flexion **(B)** protocols**.** Peak elbow angles for each coactivation level are presented, maintaining the same color scheme as Figure 2. The extension and flexion tasks follow the same pattern, but the angles are in opposite directions. The error bars show the standard error of the mean (SEM) across participants. There is more variation (larger error bars) at lower coactivation levels by approximately ±1 degrees at 0.5x compared to ±0.5 degrees at 16x.

Figure 3 lists the peak elbow angle values recorded across all conditions to see a clearer trend from Figure 2. Both Figures 3A and 3B show the peak elbow angle reaching closer to zero degrees with greater coactivation. The overall trend of Figures 2 and 3 is that peak displacements scale with the level of muscle coactivation, so the arm moves less from its starting position. The relationship between coactivation and peak elbow angle follows a clear exponential trend (Table 1), with movement decreasing as coactivation increases. The coefficient of determination or the r-squared value of 0.86 for the flexion average elbow angle, means the model explained 86% of the variance (Table 1). This indicates that the regression line approximates the actual data, which shows a strong trend in Figures 2B and 3B. However, Table 1 showed that for the extension average elbow angle, there was a lower fit ($r^2 = 0.61$), meaning coactivation explained less of the variability in movement, a weak trend in Figures 2A and 3A.

Table 1. Exponential Model Fit Results across Various Conditions and Coactivation Levels (0.5, 1, 2, and 4x)

Condition	r-squared	SSE	a	b	CI
Flexion Average Elbow Angle	0.86	0.04	2.62	-0.05	(1.93, 3.30)
Average Movement Time	0.80	2011	448.20	-0.05	(292, 604.3)
Extension Average Elbow Angle	0.61	4.45	-14.16	-0.05	(-21.43, -6.90)
Error Rate	0.78	201.65	59.07	-0.19	(-5.71, 123.80)
Metabolic Cost	0.85	14.95	117.34	0.02	(105.71,128.99)

Note. SSE = Sum of Squared Errors; CI = Confidence Interval.

a, and b are the variables in the general exponential model (equation 1).

Coactivation levels 8x and 16x were excluded.

Table 1 shows the exponential model fit which showed a strong relationship between coactivation levels and kinematics. Additionally, movement time was measured to determine the duration required for participants to return the cursor to the target following the disturbance.

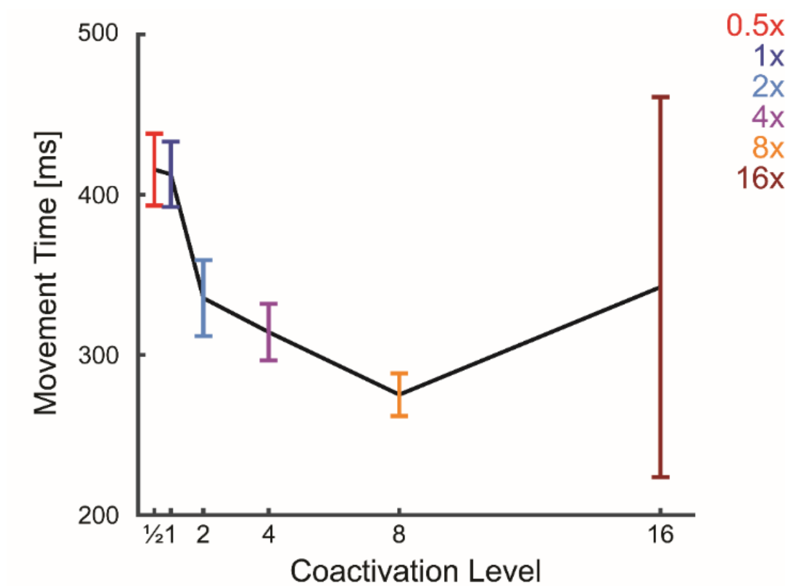

Figure 4. Average movement time for participants at different coactivation levels, with colors consistent with prior figures. The error bars show the SEM across participants.

Figure 4 shows how movement time decreases as muscle coactivation level increases, reaching the lowest point at the 8×. The recorded times are approximately 420 ms at 0.5x and 280 ms at 8x (Figure 4). At the highest level of coactivation, Figure 4 shows how the movement time increases again to about 340 ms. Figure 4 illustrates that the largest reduction in movement time occurs between the lowest and intermediate coactivation levels. Figure 4 shows that at the highest level of coactivation, it shows the most variation in movement time (approximately ±250 ms). The exponential model fit showed a decay, which means that movement time decreased with higher levels of coactivation (Table 1). This decay is consistent with the trend in Figure 4. The average movement time also had a good fit ($r^2 = 0.80$) showing a strong link between coactivation and how fast participants moved (Table 1). The overall trend was that muscle coactivation facilitates faster movements.

Muscle Activity Responses Across Coactivation Levels

The EMG data were gathered to assess muscle activity during the extension and flexion tasks across coactivation levels. Four figures present these measurements, two for each task, showing the responses of four muscles.

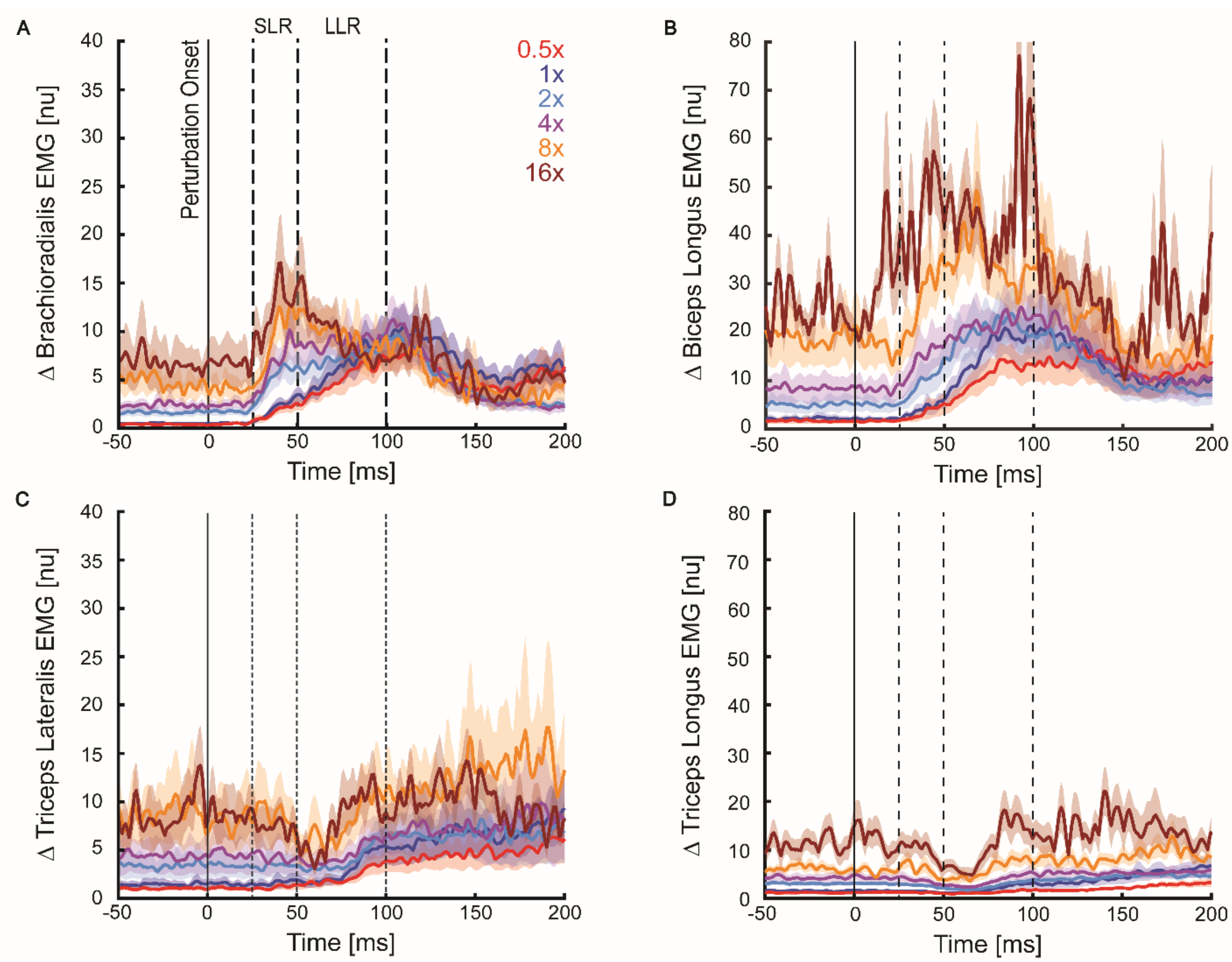

Figure 5. Group-averaged, normalized EMG activity during the extension task is shown for four muscles across different levels of coactivation, with color coding consistent with previous figures. In this task, the brachioradialis **(A)** and biceps longus **(B)** serve as the agonist muscles, while the triceps lateralis **(C)** and triceps longus **(D)** act as the antagonists. Perturbation onset occurs at 0 ms (the start of the trial). SLR is shown by the dotted line between 25-50 ms and LLR is shown by the dotted line between 50 – 100 ms. The shaded areas show the standard error of the trace across participants.

When coactivation levels increased, the agonist muscles had a stronger reaction, shown by higher EMG activity (Figure 5A and 5B). The biceps longus showed the biggest increase in activation

(Figure 5B), while the brachioradialis also became active (Figure 5A), peaking early in the movement. On the other hand, Figures 5C and 5D show the antagonist muscles were more inhibited as coactivation increased. Both the triceps lateralis (Figure 5C) and triceps longus (Figure 5D) had lower EMG activity, meaning they were more effectively suppressed. Overall, higher muscle coactivation increased agonist activity while suppressing antagonist muscles.

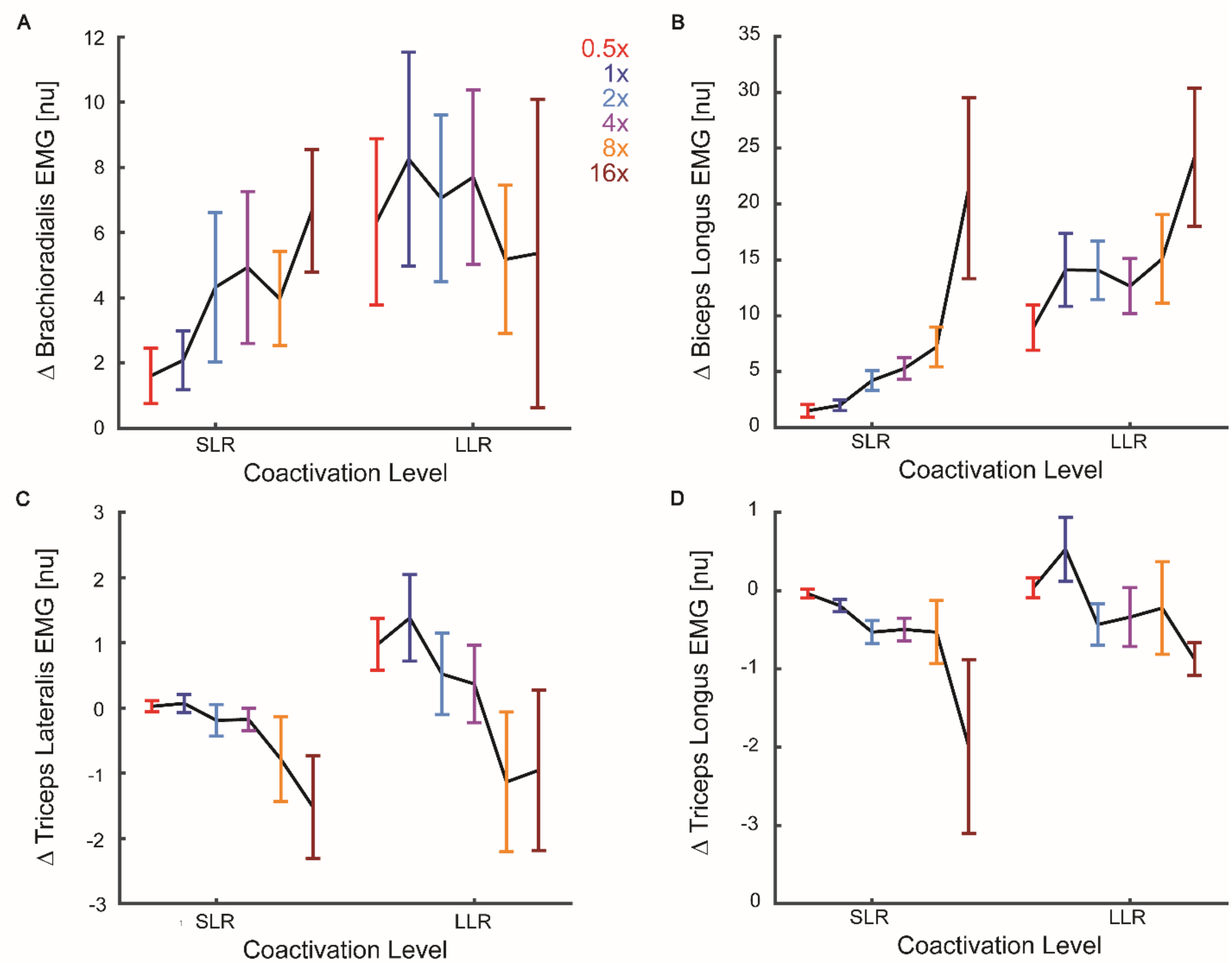

Figure 6. Summary of muscle activity for the SLR (25-50 ms) and LLR (50-100 ms) epochs across conditions for the extension task, using the same color scheme as previous figures. In this task, the brachioradialis **(A)** and biceps longus **(B)** serve as the agonist muscles, while the triceps lateralis **(C)** and triceps longus **(D)** act as the antagonists. SLR is shown on the left and LLR is shown on the right. The error bars show the SEM across participants.

Figures 6A and 6B show that during an extension task, the agonist muscles showed increased activation as the movement progressed from the SLR to the LLR, with the strongest activation seen in the LLR phase. For brachioradialis (Figure 6A), activity is approximately 1.5 nu at 0.5x and 6.5 nu at 16x in SLR, and 6.2 nu at 0.5x and 5.5 nu at 16x in LLR. Similarly, for biceps longus (Figure 6B), the activity is about 2 nu at 0.5x and 20 nu at 16x in SLR and 8 nu at 0.5x and 22.5 nu at 16x in LLR. In contrast, the antagonist muscles (Figures 6C and 6D) showed a decrease in activation during the SLR phase, which went from 0 nu to -1.5 nu for triceps lateralis (Figure 6C) and 0 nu to -2 nu for triceps longus (Figure 6D) from 0.5x to 16x coactivation. This displays that the antagonist muscles were inhibited early in the movement. Figures 6C and 6D activity slightly increased in the LLR phase for 0.5x and self-selected levels, but more coactivation resulted in the LLR phase stretch activity to drop (-1 nu for triceps lateralis and -0.8 nu for triceps longus at 16 x). The overall trend remained consistent, where the agonist muscles became more active over time, while antagonist muscles were suppressed, especially in the early phase (SLR) of the response.

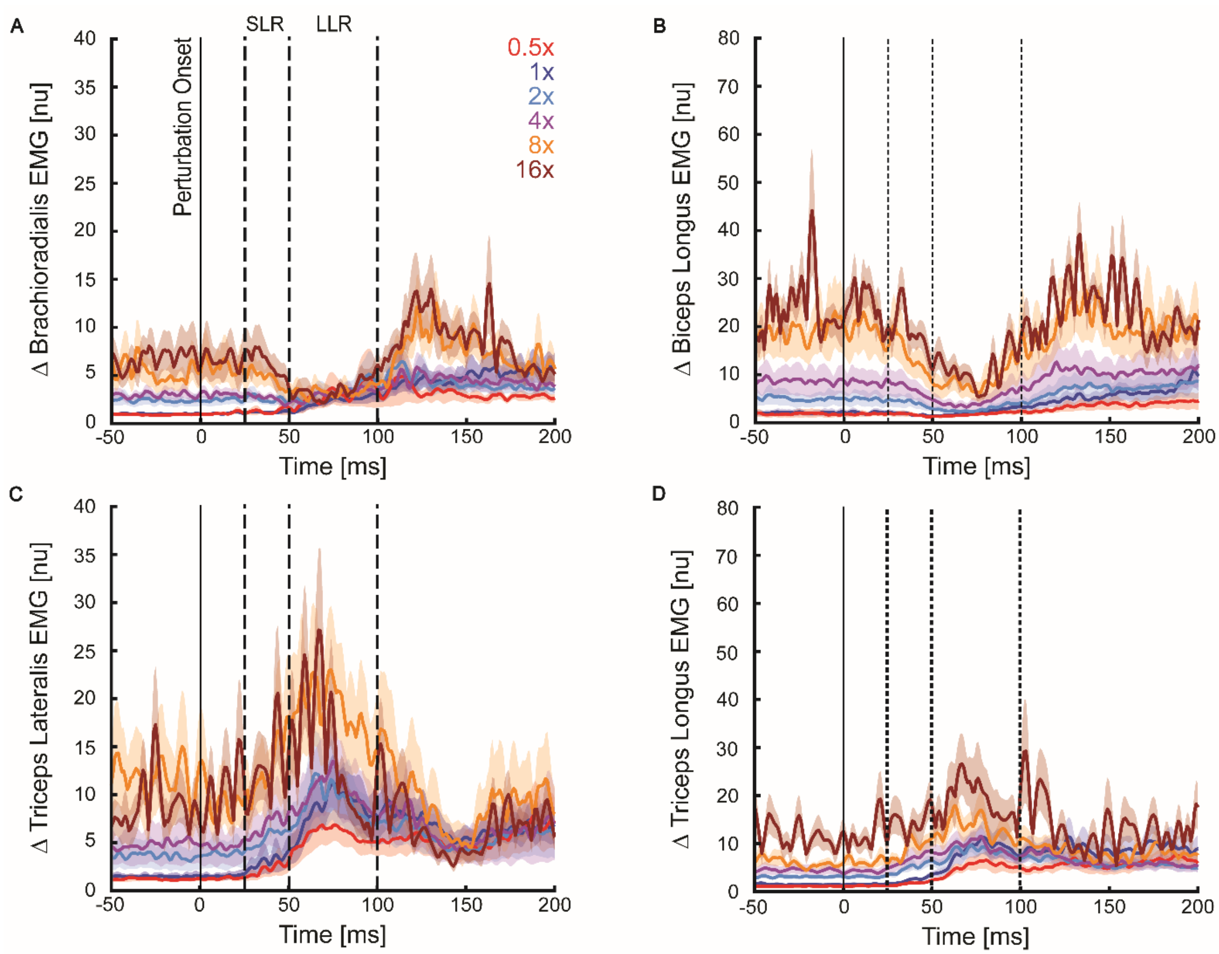

Figure 7. Group-averaged, normalized EMG activity during the flexion task is shown for four muscles across different levels of coactivation, with color coding consistent with previous figures. In this task, the brachioradialis **(A)** and biceps longus **(B)** serve as the antagonist muscles, while the triceps lateralis **(C)** and triceps longus **(D)** act as the agonist. Perturbation onset occurs at 0 ms (the start of the trial). SLR is shown by the dotted line between 25-50 ms and LLR is shown by the dotted line between 50 – 100 ms. The shaded areas show the standard error of the trace across participants.

Figure 7 shows how increasing muscle coactivation affects the strength of muscle responses to flexion perturbations. The agonist muscles (Figures 7C and 7D) show greater activation as coactivation increases. These muscles react more strongly after the perturbation onset, particularly during SLR and LLR epochs. For triceps lateralis (Figure 7C) and triceps longus (Figure 7D), stretch activity is lowest at 0.5x and highest at 16x. The triceps lateralis showed a stronger stretch activity compared to the triceps longus. In contrast, the antagonist muscles (Figure 7A and 7B) show more inhibition as coactivation increases, meaning they are less active when opposing movement. The overall trend remains consistent, showing that higher coactivation results in greater agonist activity and more suppression of the antagonist muscle.

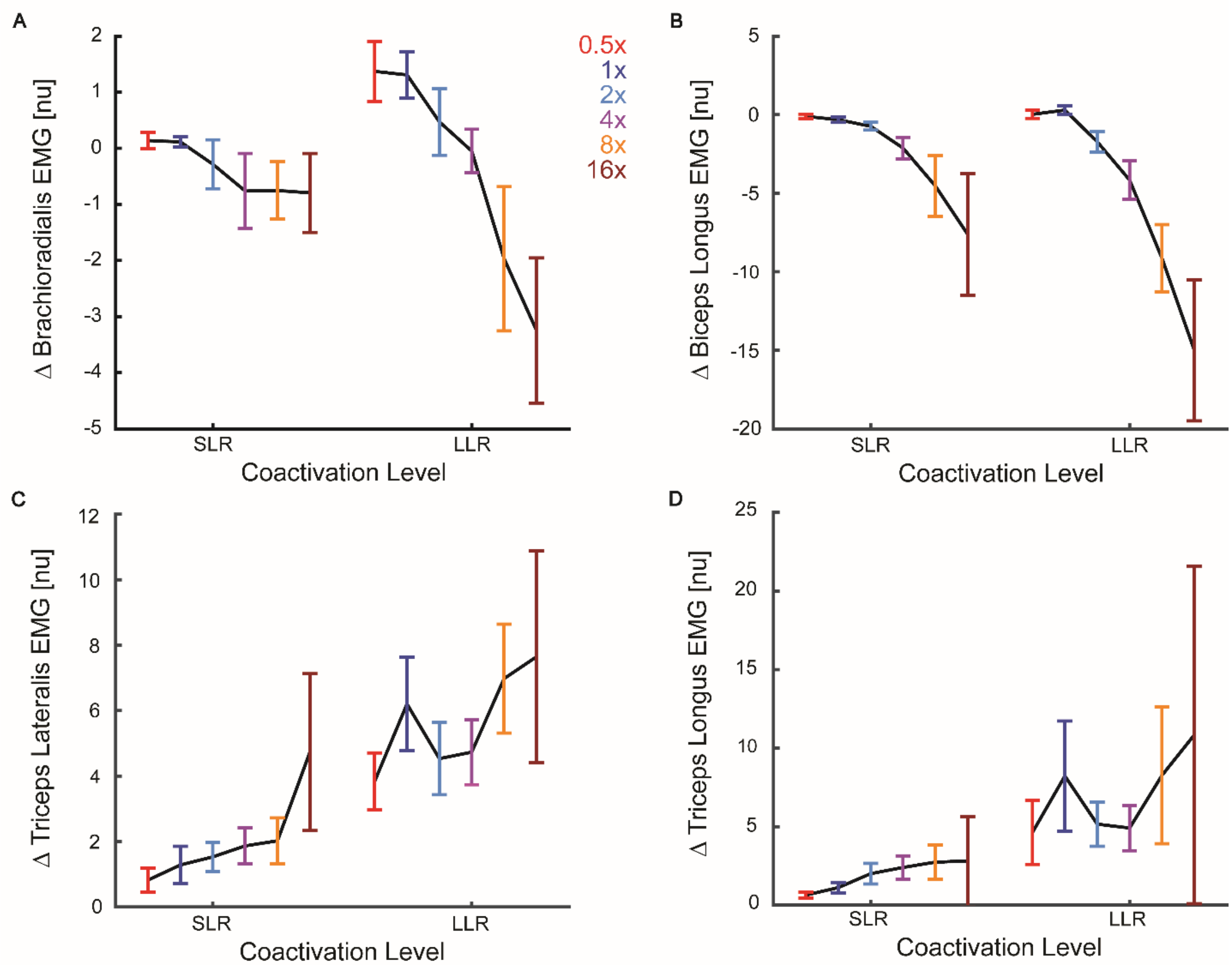

Figure 8. Summary of muscle EMG activity for the SLR (25-50 ms) and LLR (50-100 ms) epochs across conditions for the flexion task, maintaining the same color scheme as previous figures. During the flexion task, the brachioradialis **(A)** and biceps longus **(B)** act as the antagonist muscles, while the triceps lateralis **(C)** and triceps longus **(D)** act as agonist muscles. The error bars show the SEM across participants.

During the flexion task, the agonist muscles (Figures 8C and 8D) showed increased activation as coactivation levels increased. Their activity was low during SLR but became stronger in the LLR, with the triceps longus showing the greatest increase. For triceps lateralis (Figure 8C), the stretch activity is approximately 1 nu at 0.5x and 4.5 nu at 16x in SLR and 4 nu at 0.5x and 7.5 nu at 16x in LLR. Similarly, for triceps longus (Figure 8D), the activity is about 1 nu at 0.5x and 3 nu at 16x in SLR and 5 nu at 0.5x and 10 nu at 16x in LLR. In contrast, the antagonist muscles (Figure 8A and 8B) were more inhibited as coactivation increased. Their activity decreased from the SLR to the LLR phase, meaning they contributed less to resisting the flexion movement. The brachioradialis (Figure 8A) went from about 0 nu at 0.5x to -1 nu at 16x in SLR and 1.5 nu at 0.5x to -3 nu at 16x in LLR. Similarly, for biceps longus (Figure 8B), the stretch activity went from 0 nu at 0.5x to -7 nu at 16x in SLR and 0 nu at 0.5x to -15 nu at 16x in LLR. The 16x coactivation level had the largest error bars, but the overall trend remained consistent, where the agonist muscle became more active with higher coactivation levels, while antagonist muscles were more suppressed, especially in the later phase (LLR) of the response.

Metabolic Cost and Performance Across Coactivation Levels

Error rates and metabolic cost were measured to assess performance and energy expenditure across coactivation levels.

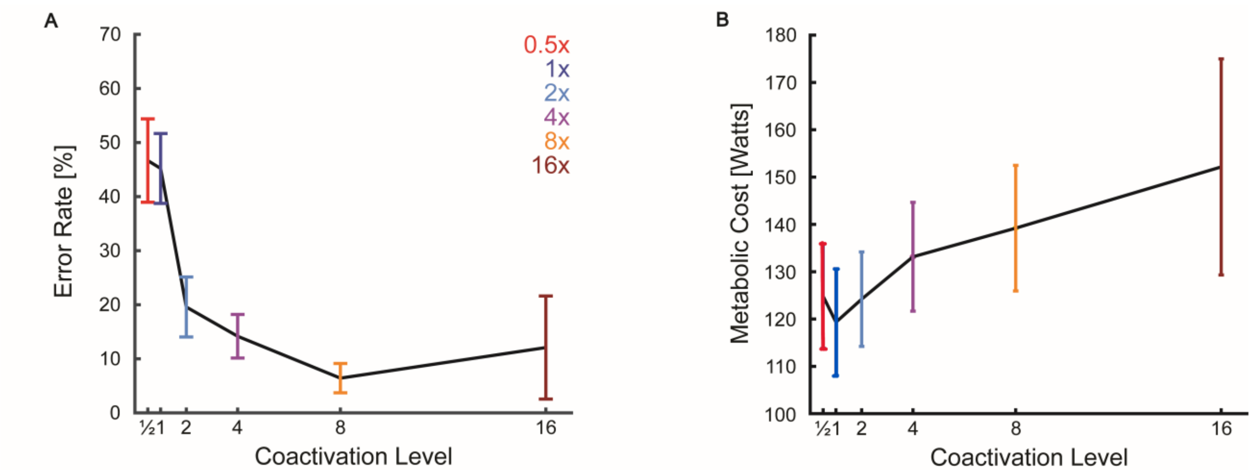

Figure 9. Average error rate **(A)** and metabolic cost **(B)** across different levels of coactivation, with color coding consistent with previous figures. The color scheme was same as previous figures. The error bars show the SEM across participants.

Figure 9A shows how error rates were highest (low performance) at the lowest coactivation levels, with approximately 48% error at 0.5x and around 46% at 1x. As coactivation increased, error rates dropped significantly, reaching about 15% at 4x and the lowest error rate of approximately 8% at 8x. However, at the highest coactivation level (16x), error rates increased again to around 13%, showing a non-linear trend (Figure 9A). The error bars for Figure 9A also showed a similar pattern, starting with large error bars at a low coactivation level and then getting smaller till 8x, while at the highest coactivation level, the error bars are the largest (approximately ±18%). The overall pattern of Figure 9A suggests that moderate to high muscle coactivation improved task performance (reduces error rate) in disturbance response before declining at extreme levels.

Although higher coactivation levels led to improved performance, it also imposed an increased metabolic cost. Figure 9B shows how higher muscle coactivation levels lead to increased metabolic costs. At the lowest coactivation levels (0.5x), the metabolic cost was about 125 W, while at the self-selected level (1x) the metabolic cost was lower, averaging about 120 W. However, as coactivation further increases, the metabolic cost rose steadily, reaching approximately 125 W at 2x, 133 W at 4x, 140 W at 8x and 150 W at 16x. At spontaneous coactivation levels, healthy participants also tend to prioritize minimizing metabolic costs, often at the expense of performance. The overall trend suggests a trade-off, that while increasing coactivation improves performance, it also results in higher energy expenditure.

In addition, the exponential model (Table 1) shows that the error rate had an r-squared of 0.78, consistent with the trend seen in Figure 9A, where errors dropped with moderate coactivation but increased at the highest levels. The error rate model calculated an ‘a’ value of 59.07 and 'b’ value of -0.19 (Table 1), which represents the sharp decline in errors with increasing coactivation. Finally, the metabolic cost had a strong r-squared value of 0.85 (Table 1), meaning the model explained 85% of the variance, which is consistent with the trend seen in Figure 9B.

Discussion

Study Overview and Broad Findings

This study questions the traditional view of coactivation as a purely mechanical strategy. It proposes that coactivation helps the nervous system with processing sensory information and controlling movement. An understanding of how the nervous system balances metabolic costs and performance provides insights into the basic mechanisms underlying motor control. The main idea was that higher muscle coactivation would improve performance but increase energy costs, while lower levels save energy but hurt performance. The results support this idea, showing that while spontaneous coactivation (1x) helped save energy, increasing muscle coactivation using biofeedback improved performance – at the expense of metabolic cost. This suggests that coactivation helps prime the nervous system to respond to sensory feedback.

Elbow Angle Kinematics

When the arm faces mechanical disturbances, there is a systemic decrease in elbow angle with higher muscle coactivation (Figures 2 and 3). This suggests that coactivation plays an important role in neural responses to feedback, helping to reduce the displacement of the arm when countering the same mechanical perturbations. This fits with Maurus et al. (2024), who argue coactivation strengthens feedback control to keep limbs steady, pushing against the old belief that muscle coactivation only stiffens muscles to resist force. Our results support the idea that coactivation helps fine-tune motor responses by limiting excessive motion. The strong trend in flexion, with an r^2 value of 0.86, shows coactivation explains 86% of the movement variance, while the weaker fit in extension with an r^2 value of 0.61, suggests that other factors (e.g., fatigue) can influence variability. This highlights coactivation's reliable role in controlling flexion but less consistency in extension. The decrease in movement with higher coactivation indicates that participants were able to change their stiffness to maintain better control, likely driven by neural feedback responses adjusting motor output (Maurus et al., 2024). This is important because it provides information to understand how coactivation balances performance and metabolic cost under disturbances.

Movement Time Optimisation

Muscle coactivation facilitates a faster return back to the target. The trend in Figure 4 and exponential model fit ($r^2 = 0.80$) shows a strong relationship between the variables and indicates that increased coactivation generally reduces movement time after perturbation. This pattern builds on Maurus et al. (2023), who state coactivation helps the nervous system react faster by increasing sensory signal processing. On the contrary, figure 4 shows that too much coactivation (16x) seems to slow things down. This could also be explained because the error bars are the widest at 16x (±250 ms), which suggests there's a lot of variation at that level, likely due to the small number of participants that completed the 16x condition. The lowest movement time at 8x hints at an ideal spot, which could really help when training for fast recovery tasks. It brings up the interesting balance between speed and energy use when it comes to coactivation, though more data is needed to confirm these trends.

Muscle Activity Patterns in Extension and Flexion Tasks

Figures 5–8 show that as coactivation increases, agonist muscle activity goes up while antagonist inhibition strengthens, more noticeably during the LLR window than the SLR window. In extension (Figures 5 and 6), the brachioradialis and biceps longus (agonists) show increased activity, while triceps lateralis and longus (antagonists) face greater inhibition. Similarly, in

Figures 7 and 8, you can see that as coactivation goes up, the activity in the triceps lateralis and triceps longus (the agonist muscles) also increases, while the antagonist muscle had the most inhibition during LLR, which might reflect the physiological delay in antagonist relaxation, possibly due to cortical feedback adjustments, as suggested by Maurus et al. (2024). Greater inhibition of antagonist muscles suggests improved reciprocal inhibition, reducing opposition to the desired movement (Geertsen et al., 2011). The overall trend was that coactivation boosts agonist activity while reducing antagonist engagement, which supports Cluff and Scott's (2013) idea that coactivation can help with limb stability. They also challenge the old view that coactivation is just about resisting movement, suggesting it plays a role in enhancing feedback from the nerves. This builds on Molina-Rueda et al. (2023), linking coactivation to stability but suggesting that we should also think about feedback adjustment. The findings imply that greater coactivation helps produce more force while also adjusting reflexes, which could lead to better stability and precision during extension and flexion tasks.

Performance and Metabolic Cost Trade-Off

The trends in Figure 9 show a clear trade-off between performance (decrease in error rate) and metabolic cost as muscle coactivation increases. The figure plots error rates dropping from 48% at 0.5x to 8% at 8x, then rising to 13% at 16x, and metabolic cost climbing from 120 to 150 Watts. This aligns with Peterson and Martin (2010), who note coactivation improves stability but raises energy use yet challenges the idea that less coactivation always conserves energy for mobility. The higher metabolic cost at 0.5x (125 Watts) compared to self-selected 1x (120 Watts) might stem from compensatory muscle recruitment to maintain stability, as Kistemaker et al. (2006) suggest, requiring extra effort below the natural efficiency point. This finding underscores the physiological trade-off between stability and metabolic efficiency, which is crucial for understanding motor control strategies in both everyday movements and specialized tasks. In addition, at the highest coactivation level, the error bars for error rate were the largest, approximately ±18% (Figure 9A), which suggests that there is a lot of variation at that level, likely due to the small number of samples that completed the 16x condition. However, the overall trend suggests that as muscle coactivation increases, error rates decrease, but metabolic cost increases, which links back to the question of how coactivation trades off performance and metabolic cost.

Strengths of the Study

This study stands out because it uses biofeedback to increase the muscle coactivation, showing trends that it does more than just make muscles stiff. By looking at the EMG data, we split SLR and LLR to show that coactivation really helps with neural feedback, especially in the LLR window. This pushes against the old idea that coactivation only fights motion. It's a big deal because it gives solid proof that coactivation fine-tunes stability and precision, helping us understand how the nervous system improves performance in everyday movements. Our study shows that there is a complex interplay between metabolic cost and performance in the control of human arm movements and postures.

Limitations of the Study

The small group of 12 participants limits generalizability across diverse populations. Some participants could not complete every condition, leading to less reliable data, with wider error bars and weaker r-squared values in certain measures. We did not track fatigue or prior exercise, which could affect muscle activity and energy use results. The study also overlooked how sex, age, or body size might shape responses to disturbances or energy demands, possibly altering coactivation's role in movement control. While increased EMG responses to perturbations show coactivation's sensory role, as Maurus et al. (2023) suggest, we could not identify where neural processing occurs, such as in the primary motor cortex or cerebellum. The exclusion of 8x and 16x conditions in Table 1, due to limited trials, also restricts insights into high coactivation effects.

Future Implication and Direction

This project not only expands our understanding of human movement but also broadens the scope of future research and real-life applications in rehabilitation and sports science. Individuals with motor deficits, such as stroke survivors or those with multiple sclerosis, often adopt increased coactivation levels as a compensatory technique (Molina-Rueda et al., 2023). Since our findings found that coactivation improves sensory feedback, rehabilitation efforts may shift toward optimizing sensory processing rather than simply increasing muscle stiffness. To understand the range of scenarios in which coactivation can be useful, research should focus on a larger population, which includes older adults, athletes, and persons with impairments. Also, looking into how coactivation relates to things like muscle fatigue can help us understand motor control better.

Conclusion

We aimed to figure out how muscle coactivation shows a trade-off between performance and metabolic cost in upper limb posture control with mechanical disturbances. Our findings show that at spontaneous coactivation levels, participants tend to prioritize reducing the metabolic cost at the expense of performance. However, increasing coactivation through biofeedback improves performance but also raises the metabolic cost. This emphasizes the role of muscle coactivation in balancing sensory feedback. In summary, our preliminary results suggest a complex relationship between metabolic cost and performance in controlling arm movements and postures. This research provides a new framework for understanding why the nervous system coactivates skeletal muscles, which may allow for better treatments in a clinical setting for excessive muscle tone (muscle spasticity) and abnormally sensitive reactions to sensory feedback (hyperreflexia).

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Author contributions

Conceptualization, P.M., and T.C.; Methodology, B.D., and T.C.; Software, P.M., and T.C.; Formal Analysis, B.D., J.R., P.M., and T.C.; Investigation, B.D., and P.M.; Writing – Original Draft, B.D.; Writing – Review & Editing, B.D., and T.C.; Funding Acquisition, T.C.; Supervision, T.C.

Declaration of interests

The authors declare no competing interests.

Inclusion and diversity

We support inclusive, diverse, and equitable conduct of research.

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